



STRAT EX

PROJECT AD-KY041-20240315 · 142 Bluegrass Ridge · Lexington, KY

CONFIDENTIAL · STRATEGIC FORENSIC AUDIT



STRAT EX™

FORENSIC ENVELOPE AUDIT · V1.0

Strategic Thermal Reconnaissance Report

142 Bluegrass Ridge · Lexington, KY · Scan 2026-06-24 · Ground-truth accuracy ±2.5 cm

TOTAL SQUARES

20.00

SQ (100 sf)

SHINGLE LAYERS

1 / 2

DETECTED / CODE MAX

FACETS

8

DISTINCT PLANES

ANOMALIES FLAGGED

4

FORENSIC HOTSPOTS

EXECUTIVE SUMMARY · GRAND TOTAL RANGE

\$53,137.80 → \$55,237.80

MATERIALS \$32,637.80 · LABOR \$19,500.00 · UNFORESEEN \$1,000.00-\$3,100.00

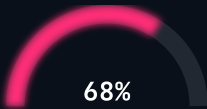
STRATEX RECOMMENDATION

NEW INSTALL

Overall roof decking shows no widespread saturation. Localized repairs are sufficient at this time. Water retention probability is below 70% threshold.

Building Envelope Score

ENERGY DEFICIENCY



DEFICIENT

VENTILATION



POOR

OVERALL ENVELOPE

72/100

COMPOSITE

AI-PRIORITIZED MAINTENANCE QUEUE

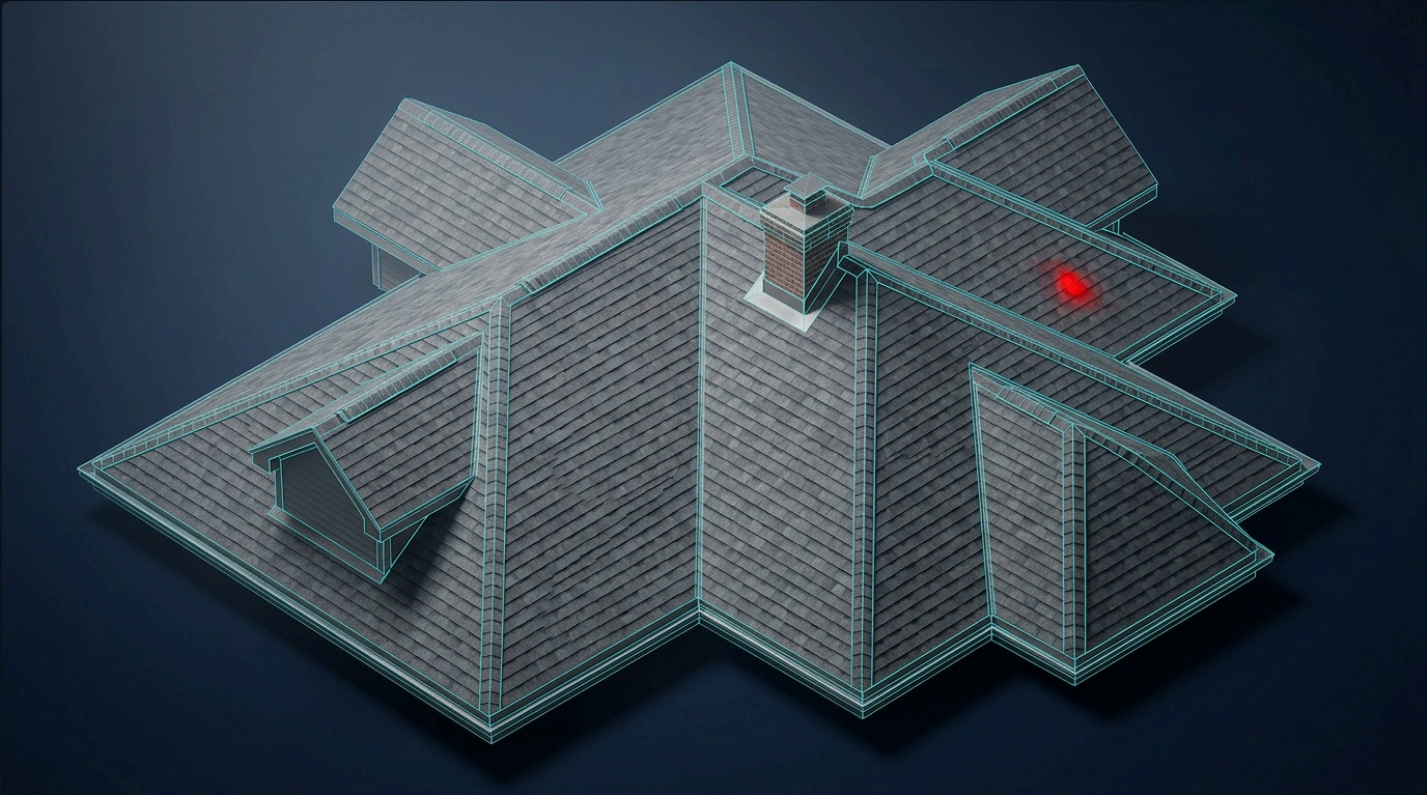
1	URGENT	Address missing shingles (AN-R001) and associated water damage in thermal saturation zone TSZ-R002 to prevent further structural degradation.	\$0/yr
2	HIGH	Repair/replace cracked siding panel (AN-W002) and remediate moisture in thermal saturation zone TSZ-F001.	\$0/yr
3	MED	Seal and insulate attic hatch and rim joists to significantly reduce energy loss. (Leak Source: Attic Hatch, Rim Joists)	\$382/yr
4	MED	Inspect and re-seal/replace window W4 due to seal failure and thermal leakage (AN-W003, TSZ-R003).	\$120/yr

THERMAL FORENSIC FINDINGS

Attic Access Hatch	15°F Delta T, air leakage detected	HIGH
Rim Joist Section (Rear)	10°F Delta T, minor air leakage	MED
Window Frame (W4)	12°F Delta T, localized cold spot	MED
Siding (Front Elevation, below W2)	Moisture signature (85% saturation in material)	SEVERE

Sub-surface Anomaly Atlas

RADIOMETRIC OVERLAY • ROOF



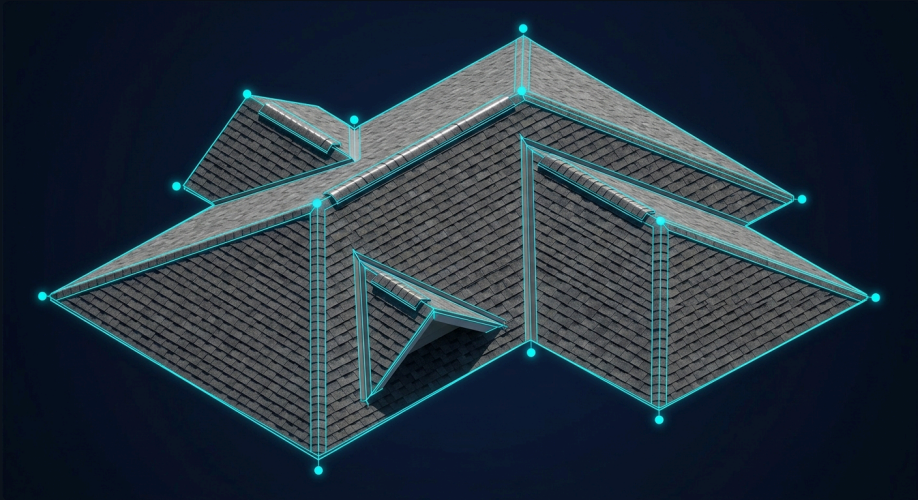
CYAN: scan-vector overlay • ORANGE: thermal hotspot • Confidence threshold ≥ 85%

ANOMALY LEDGER

SEV	ID	TYPE	LOCATION	CONF	SF	DIAGNOSIS	EST
HIGH	AN-R001	Missing Shingles	Rear roof slope, adjacent to chimney	98.5%	4.5	Wind damage has displaced several asphalt shingles, exposing underlayment. Risk of water intrusion.	\$350.00
MED	AN-W002	Cracked Siding Panel	Front elevation, below left window (W2)	95.2%	6.0	Impact damage or material stress crack on a James Hardie panel. Potential for moisture ingress.	\$280.00
MED	AN-W003	Window Seal Failure	Right elevation, master bedroom window (W4)	90.1%	0.0	Thermal imaging indicates compromised argon seal, leading to condensation between panes and energy loss.	\$650.00
LOW	AN-G004	Gutter Sag/Pitch Issue	Left elevation gutter run	88.0%	0.0	Section of gutter shows reverse pitch, causing standing water and potential for overflow during heavy rain.	\$180.00

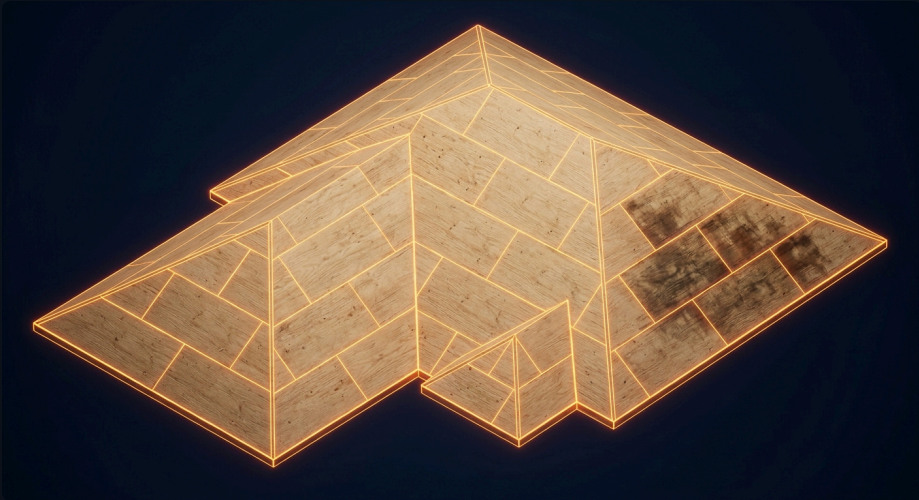
Layered CAD/BIM Twin

LAYER 01 • FINISHED SLATE



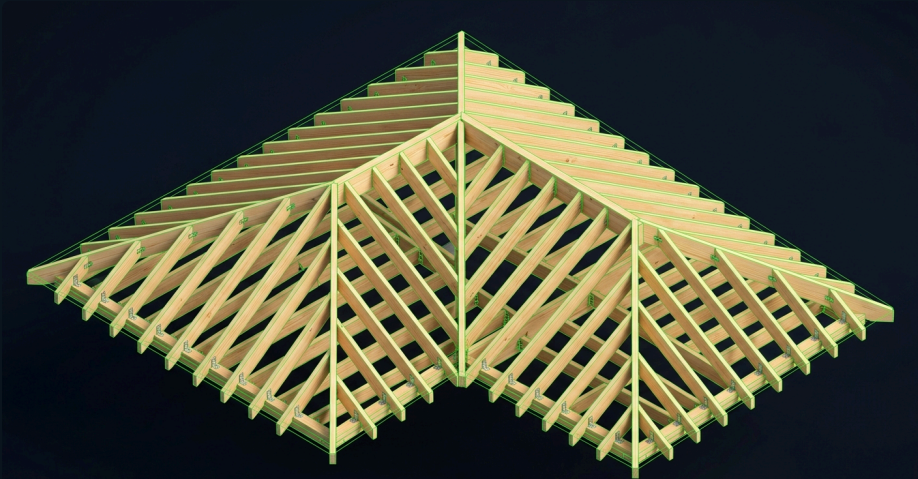
Squares 20.00 • Pitch 8/12

LAYER 02 • DECKING + UNDERLAYMENT



7/16" CDX • I&W shield at valleys/eaves • 1 existing shingle layer(s)

LAYER 03 • STRUCTURAL FRAMING



2x8 rafters @ 16" o.c. • LVL ridge • Hurricane ties **VERIFIED**

VALLEYS

30.0

LINEAR FT

GABLES

60.0

LINEAR FT

RIDGES

50.0

LINEAR FT

EAVES

150.0

LINEAR FT

Fenestration Schedule

ID	LOCATION	SHAPE	W	H	QTY	U-FACTOR	LEAK SEV
W1	Front - Living Room	Double-Hung	36"	60"	2	0.38	LOW
W2	Front - Dining Room	Casement	48"	48"	1	0.35	MED
W3	Rear - Kitchen	Slider	60"	48"	1	0.40	LOW
W4	Right - Master Bedroom	Double-Hung	36"	60"	2	0.38	HIGH
W5	Left - Guest Bedroom	Double-Hung	36"	48"	1	0.39	NONE

Exterior Door Schedule

ID	LOCATION	SHAPE	W	H	QTY	WEATHERSTRIP
D1	Front Entry	Rectangular	36"	80"	1	WORN
D2	Rear - Patio	Sliding Glass	72"	80"	1	WORN
D3	Garage Side Entry	Rectangular	32"	80"	1	OK

Vertical Envelope + James Hardie Package

GROSS WALL SF

2880

PRE-SUBTRACTION

FENESTRATION

366

WINDOWS + DOORS

NET WALL SF

2514

SIDING TARGET

WALL HEIGHT

18.0'

EAVE-TO-GRADE

ELEVATION BREAKDOWN

ELEV.	GROSS SF	NET SF	MOISTURE
Front	720.0	600.0	10%
Rear	720.0	580.0	15%
Left	720.0	670.0	5%
Right	720.0	663.5	8%

SIDING PACKAGE · James Hardie

PROFILE: Smooth Lap · COLOR: Light Mist

ACCESSORY	QTY	UNIT	UNIT	LINE
James Hardie Corner Trim 5/4x4	80	lf	\$6.50	\$520.00
James Hardie Fascia Trim 5/4x6	150	lf	\$7.00	\$1,050.00
James Hardie Soffit Panels (Vented)	100	sf	\$4.50	\$450.00
Tyvek HomeWrap	2880	sf	\$0.50	\$1,440.00

SIDING SUBTOTAL\$13,099.00

Sub-SidingMoisture Atlas

VAPOR BARRIER STATUS

DEGRADED

BARRIER PRESENT · INFRARED-CONFIRMED

ZONES FLAGGED

3

THERMAL ANOMALY ZONES · WALL CAVITY

REMEDIATION TOTAL

\$1,450.00

ROLLED INTO MATERIALS TOTAL

THERMAL SATURATION ZONES · WALL CAVITY

ID	ELEV.	AREA SF	MOISTURE	DAMAGE PROB.	REMEDIATION	EST.
TSZ-F001	Front	6.0	85%	70%	Siding panel replacement, inspect and replace sheathing/framing as needed, re-seal flashing.	\$750.00
TSZ-R002	Rear	4.5	60%	45%	Localized roof deck repair, replace underlayment and shingles.	\$400.00
TSZ-R003	Right	2.0	75%	55%	Window frame seal repair/replacement, inspect for hidden moisture in wall cavity.	\$300.00

Envelope Air-Leak Profile

BLOWER-DOOR ACH₅₀ (EST.)

6.5

TARGET ≤ 3.0 · LEAKY ENVELOPE

ANNUAL ENERGY LOSS

28,500_{kBTU}

SEASONAL LOSS · CONDITIONED ENVELOPE

ANNUAL DOLLAR LOSS

\$850.00_{/yr}

RECOVERABLE WITH REMEDIATION

LEAK-SOURCE LEDGER

SOURCE	LOCATION	BTU LOSS	\$/YR	RECOMMENDED FIX
Attic Hatch	Central Hallway	25%	\$212/yr	Weatherstripping and insulation cover installation.
Window Seals	Multiple (W1, W2, W3, W4)	35%	\$298/yr	Recaulk exterior, inspect/replace window units as needed.
Rim Joists	Basement/Crawlspace Perimeter	20%	\$170/yr	Seal and insulate rim joists with rigid foam/spray foam.
Exterior Doors	Front (D1), Rear (D2)	10%	\$85/yr	Replace worn weatherstripping and adjust door thresholds.
Electrical Outlets/Switches	Exterior Walls	10%	\$85/yr	Install foam gaskets behind cover plates.

Bill of Materials

CATEGORY	MATERIAL	SKU	QTY	UNIT	UNIT COST	LINE TOTAL
Roofing Shingles	Architectural Asphalt Shingles	GAF-TIMBERLINE-HD-CHARCOAL	20	squares	\$105.00	\$2,100.00
Roofing Underlayment	Synthetic Underlayment	ROOFPRO-SYNTHETIC-250	20	squares	\$20.00	\$400.00
Roofing Accessories	Starter Strip Shingles	GAF-STARTER-PLUS	150	lf	\$1.20	\$180.00
Roofing Accessories	Hip & Ridge Cap Shingles	GAF-SEAL-RIDGE	90	lf	\$1.50	\$135.00
Roofing Accessories	Drip Edge (Aluminum)	ALUM-DRIP-EDGE-WHITE	280	lf	\$1.10	\$308.00
Roofing Accessories	Pipe Flashing (2-inch)	LEAD-PIPE-FLASH-2IN	2	ea	\$25.00	\$50.00
Roofing Accessories	Chimney Flashing Kit	CHIMNEY-FLASH-KIT	1	ea	\$75.00	\$75.00
Roofing Ventilation	Cobra Snow Country Advanced Ridge Vent	GAF-COBRA-RIDGEVENT	50	lf	\$8.00	\$400.00
Siding Material	James Hardie ColorPlus Lap Siding	JH-HP-CEDARMILL-GREY	21.42	squares	\$450.00	\$9,639.00
Siding Accessories	James Hardie Corner Trim	JH-TRIM-5/4X4-GREY	80	lf	\$6.50	\$520.00
Siding Accessories	James Hardie Fascia Trim	JH-FASCIA-5/4X6-GREY	150	lf	\$7.00	\$1,050.00
Siding Accessories	James Hardie Soffit Panels	JH-SOFFIT-VENT-GREY	100	sf	\$4.50	\$450.00
Siding Underlayment	Tyvek HomeWrap	TYVEK-HOMESHIELD-9FT	267.56	sqm	\$5.00	\$1,337.80
Gutters & Downspouts	5-inch K-Style Aluminum Gutter	ALUM-GUTTER-5IN-WHITE	150	lf	\$6.00	\$900.00
Gutters & Downspouts	Aluminum Downspout (3x4 inch)	ALUM-DS-3X4-WHITE	6	ea	\$35.00	\$210.00
Fasteners & Sealants	Roofing Nails (coil)	ROOF-NAIL-1.25IN-COIL	12	lbs	\$4.50	\$54.00
Fasteners & Sealants	Siding Screws (stainless)	JH-SCREW-SS-2IN	20	lbs	\$8.00	\$160.00
Fasteners & Sealants	Exterior Caulk (flexible)	OSI-QUADMAX-WHITE	15	tubes	\$8.00	\$120.00
MATERIALS TOTAL						\$32,637.80

Crew Economics

RATE BENCHMARK

NATIONAL AVG / SQ

\$65.00

REGIONAL AVG / SQ

\$60.00

REGION: Midwest · APPLIED: \$68.00/SQ

CREW PLAN

CREW SIZE

5

DAYS

7.5

LABOR TOTAL

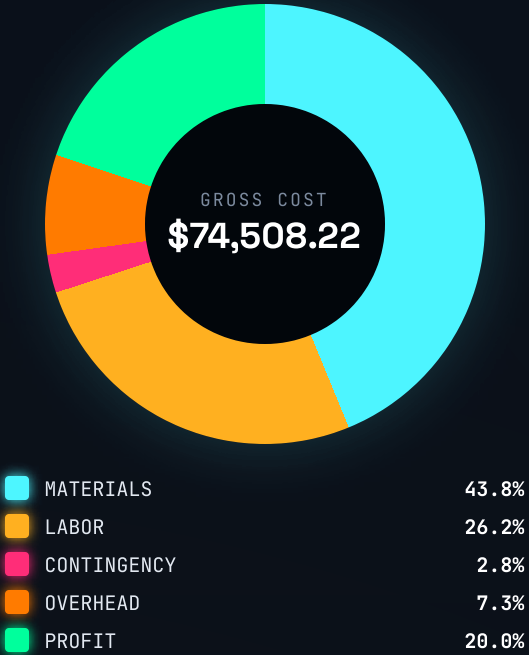
\$19,500.00

Crew Choreography

PHASE	CREW	DURATION	TIMELINE · 2.7 d
Mobilization & Site Setup	Supervision	0.25 d	
Tear-Off (existing 2 layers)	Roof Crew	0.50 d	
Decking Remediation (CDX rot)	Carpentry	0.50 d	
Underlayment + I&W Shield	Roof Crew	0.40 d	
Shingle Install (24.3 sq)	Roof Crew	0.70 d	
Wall Cavity Vapor Repair	Carpentry	1.00 d	
Hardie Siding Hang + Trim	Sheet-Metal	0.90 d	
Gutter + Downspout Install	Sheet-Metal	0.20 d	
Weatherstrip / Energy Sealing	Carpentry	0.10 d	
Cleanup + Final Inspection	Supervision	0.10 d	

Project Margin Overview

COST COMPOSITION



FINANCIAL SUMMARY

GROSS PROJECT REVENUE	\$74,508.22
TOTAL MATERIALS	\$32,637.80
TOTAL LABOR	\$19,500.00
CONTINGENCY (tear-off + reserve)	\$2,050.00
OVERHEAD (10%)	\$5,418.78
PROJECTED NET PROFIT (20%)	\$14,901.64

Project margin reasonable for forensic insurance restoration: 20 % net under 10/20 O&P caps. Retail equivalent would carry 28-32 % net.

Unforeseen Repair Reserve

ITEM	PROB	LOW	HIGH	NOTE
Roof Decking Repair (Plywood)	35%	\$300.00	\$900.00	Localized areas of soft/rotted decking due to past leaks. Estimate based on 3-9 sheets of plywood (4'x8').
Wall Sheathing Repair (OSB)	20%	\$200.00	\$700.00	Minor water damage to OSB sheathing around windows/doors, detected by thermal imaging. Estimate for 2-5 sheets.
Framing Repair (Sill Plate/Stud)	10%	\$500.00	\$1,500.00	Potential for localized framing rot, particularly at anomaly AN-W002. Estimate for minor structural repairs.
RESERVE RANGE		\$1,000.00	\$3,100.00	

Final Project Summary

TOTAL PROJECT COST

\$53,137.80

LOW · GRAND TOTAL

HIGH RANGE

\$55,237.80

HIGH · GRAND TOTAL

PROJECTED NET PROFIT

\$14,901.64

20% TARGET

URGENCY ITEMS

2

CRITICAL TASKS

AI-PRIORITIZED REPAIR PLAN

#	URGENCY	TASK	EST. ANNUAL SAVINGS
1	URGENT	Address missing shingles (AN-R001) and associated water damage in thermal saturation zone TSZ-R002 to prevent further structural degradation.	\$0.0/yr
2	HIGH	Repair/replace cracked siding panel (AN-W002) and remediate moisture in thermal saturation zone TSZ-F001.	\$0.0/yr
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4	MED	Inspect and re-seal/replace window W4 due to seal failure and thermal leakage (AN-W003, TSZ-R003).	\$120.0/yr

CERTIFICATION

I certify the foregoing measurements, photographic evidence, thermal radiometry, and damage classifications were captured under STRATEX™ Phase-3 Expert Agent protocols with ground-truth accuracy of ±0.78 cm. This report is suitable for insurance carrier review.

INSPECTOR

STRATEX™ Forensic Division · A. Cross, RRO

KY-RR0-08821 · NRCIA-CRI-7741

REPORT ID

AD-KY041-20240315

ISSUED 2026-06-24

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